Nykaa Product Retention & Customer LTV Analytics

Comprehensive Business Case Study, SQL Data Pipeline, Behavioral RFM Modeling, and Strategic RICE Growth Roadmap

PRIMARY AUTHOR
Product Analytics Team

DATA SCOPE
10,000 Orders (2,451 Users)

TECH STACK
MySQL 8.0, Python, Power BI

DOMAIN
Beauty & E-Commerce Analytics

1. Executive Summary & Business Context

In beauty and personal care e-commerce, customer acquisition cost (CAC) continues to escalate due to media inflation across digital performance channels. For category leaders like Nykaa, long-term unit economics and platform profitability depend on accelerating repeat purchase velocity and expanding Customer Lifetime Value (LTV).

This executive report analyzes 10,000 transaction records across 2,451 unique buyers. By establishing month-over-month cohort decay matrices, segmenting customers via Recency-Frequency-Monetary (RFM) modeling, and profiling category replenishment cycles, this project identifies critical churn bottlenecks and formulates a prioritized growth roadmap.

10,000	2,451	75.1%	47 Days
Transactions Analyzed	Unique Buyers	Month 1 Drop-off Rate	Median Reorder Window

2. Primary Business Questions

To provide actionable insights for growth, product, and CRM marketing leadership, this study answers five foundational business questions:

- **Cohort Velocity:** At what exact post-purchase interval do buyers lapse, and where does repeat retention stabilize?
- **RFM Segmentation:** What proportion of the active user base constitutes core revenue drivers vs. at-risk shoppers?
- **Replenishment Clock:** What are the distinct reorder cycles across Skincare, Makeup, Haircare, and Personal Care?
- **Channel Efficiency:** Which acquisition and engagement channels generate the highest concentration of high-LTV Privé VIP members?
- **Feature Roadmap:** Which growth interventions yield the highest return on engineering effort using RICE scoring?

3. Step-by-Step Technical Solution & Data Pipeline

The underlying data pipeline transforms unformatted transactional logs into structured CTE analytical tables inside MySQL Workbench. Data hygiene procedures addressed string-formatted dates (e.g., `M/D/YYYY`) by applying sequential `STR_TO_DATE()` fallback functions prior to calculating cohort month gaps and windowed RFM quartiles.

```
-- STEP 1: Safe Update Bypass & Date Serialization SET SQL_SAFE_UPDATES = 0; UPDATE nykaa_retention_data SET Order_Date = COALESCE( STR_TO_DATE(Order_Date, '%c/%e/%Y %H:%i:%s'), STR_TO_DATE(Order_Date, '%c/%e/%Y'), STR_TO_DATE(Order_Date, '%Y-%m-%d %H:%i:%s') ) WHERE Order_Date NOT LIKE '202%-%;'; ALTER TABLE nykaa_retention_data MODIFY COLUMN Order_Date DATETIME; -- STEP 2: Cohort Month Gap via TIMESTAMPDIFF WITH UserFirstOrder AS ( SELECT User_ID, MIN(Order_Date) AS first_order_date, DATE_FORMAT(MIN(Order_Date), '%Y-%m-01') AS cohort_month FROM nykaa_retention_data GROUP BY User_ID ) SELECT o.User_ID, f.cohort_month, TIMESTAMPDIFF(MONTH, f.first_order_date, o.Order_Date) AS period_number FROM nykaa_retention_data o INNER JOIN UserFirstOrder f ON o.User_ID = f.User_ID;
```

4. Dataset Schema & Data Attributes

The primary table `nykaa_retention_data` stores granular order event records enriched with channel acquisition tags and user loyalty status:

Field Name	Data Type	Description	Transformation Note
`Order_ID`	VARCHAR(50)	Unique order identifier	Validated against duplicate logs
`User_ID`	VARCHAR(50)	Unique customer identifier	Key for user-level aggregations
`Order_Date`	DATETIME	Transaction timestamp	Cast from text via multi-pattern logic
`Order_Value`	DECIMAL(10,2)	Monetary order spend (INR)	Aggregated for RFM Monetary scoring
`Category`	VARCHAR(50)	Product vertical	Skincare, Makeup, Personal Care, Haircare
`Is_Prive_Member`	TINYINT(1)	Privé VIP loyalty flag	Binary classification (0/1)

> **Pipeline Integrity:** Bypassing Safe Update mode specifically during date conversion prevented schema migration blocks while maintaining strict session safety controls.

5. Cohort Retention Velocity Findings

Month-over-month cohort decay analysis reveals a steep initial drop-off immediately following a buyer's first transaction, followed by a resilient long-term active baseline.

Acquisition Cohort	Cohort Size	Period 0 (M0)	Period 1 (M1)	Period 2 (M2)	Retention Baseline
2025-01	412 Users	100.0%	24.8%	27.9%	Stabilized @ ~28%
2025-02	385 Users	100.0%	25.2%	28.1%	Stabilized @ ~28%
2025-03	430 Users	100.0%	24.2%	27.4%	Stabilized @ ~27%
2025-04	398 Users	100.0%	25.6%	28.5%	Stabilized @ ~28%
Average Benchmark	**2,451 Users**	**100.0%**	**24.9%**	**28.0%**	**28.0% Baseline**

6. Behavioral RFM Customer Segmentation

Using `NTILE(4)` window functions across Recency (days inactive), Frequency (order count), and Monetary (cumulative spend), buyers were classified into four operational tiers:

Customer Segment	User Count	% Share	Avg Orders	Avg Spend (INR)	Strategic Focus
Loyal Repeat Buyers	1,273	51.9%	5.3 orders	₹6,850.00	Core Revenue Defense
Occasional Shoppers	615	25.1%	2.8 orders	₹3,420.00	Frequency Nudges
At Risk / Churned	420	17.1%	1.8 orders	₹1,950.00	Win-Back Incentives
Champions / Privé VIPs	143	5.9%	8.2 orders	₹13,920.00	Exclusive Perks

> **Key Takeaway:** 51.9% of users belong to the Loyal Repeat Buyers segment, demonstrating that platform habituation is exceptionally strong once customers complete a second purchase.

7. Category Reorder Clock Analysis

Measuring the median inter-purchase interval across product lines reveals distinct replenishment schedules per vertical:

Category Vertical	Order Volume	Volume Share	Median Reorder Window	Primary Reorder Nudge Timing
Skincare	3,900	39.0%	47 Days	Day 42–45 Automated Push
Makeup	3,000	30.0%	52 Days	Day 48 Personalized Shade Alert
Personal Care	1,500	15.0%	48 Days	Day 45 WhatsApp Bundle Offer
Haircare & Fragrance	1,600	16.0%	60–90 Days	Day 60 Cross-Category Incentive

8. Acquisition & Loyalty Channel Conversion

Evaluating order distribution and Privé VIP penetration across traffic channels reveals clear channel specializations:

Channel Source	Total Volume	Privé VIP Member Share	Strategic Role
App Push Notifications	3,001 Orders	24.5%	High-Volume Re-engagement
Organic Search	2,513 Orders	18.2%	Top-of-Funnel Acquisition
WhatsApp Nudges	2,105 Orders	**32.3% (Highest)**	VIP & High-LTV Conversion
Social Performance Ads	2,381 Orders	12.4%	First-Time Buyer Onboarding

> **Channel Insight:** While App Push drives the highest raw transaction volume, WhatsApp Nudges exhibit the highest concentration of Privé VIPs (32.3%), making it the premier channel for premium loyalty campaigns.

9. Strategic Recommendations & RICE Framework

To address early post-purchase churn and maximize repeat velocity, proposed product interventions were evaluated using the RICE scoring formula ($\text{RICE} = [\text{Reach} \times \text{Impact} \times \text{Confidence}] / \text{Effort}$):

Initiative / Feature	Reach	Impact	Confidence	Effort	RICE Score	Priority
45-Day Automated Category Replenishment Nudges	2,000	3.0 (High)	90%	2 wks	**2,700**	**P0 (Immediate)**
Privé VIP Loyalty Gamification	2,000	2.0 (Med)	80%	4 wks	**800**	**P1**
Personalized Shade & Skin Profiler	1,000	3.0 (High)	80%	6 wks	**400**	**P2**

10. Projected Business Impact & ROI

- **Retention Uplift:** Automated 45-day replenishment alerts timed to Skincare and Makeup cycles are projected to reduce Month 1 churn by **12–15%**.
- **LTV Expansion:** Nudging Occasional Shoppers into Loyal Repeat Buyers via WhatsApp expands average order frequency from **2.8 to 5.3 orders/user**.
- **Lapsing Revenue Recovery:** Win-back flows targeting inactive users (>120 days) are estimated to recover **8–10%** of churned spend.

11. Limitations & Next Steps

Findings reflect historical transaction logs and should be validated via A/B testing prior to full platform deployment. Next steps include integrating real-time replenishment triggers into the marketing automation engine.

> **Executive Conclusion:** Synchronizing automated CRM nudges with the 47-day replenishment clock transforms early drop-offs into high-margin repeat revenue.